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Software Requirements Specification - Faculty of Technology Management System

ICT2132 Object Oriented Programming Practicum Mini Project

GROUP 14

UNIVERSITY OF RUHUNA

Bachelor of Information and Communication Technology Degree

Level II (Semester I)

ICT2132 – Object Oriented Programming Practicum

Mini Project – Faculty of Technology Management System

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the Faculty of Technology Management System (FTMS), developed as the Mini Project for the course unit ICT2132 – Object Oriented Programming Practicum at the University of Ruhuna. The purpose of the system is to digitize and centralize the academic administration processes of the Faculty of Technology, including user profile management, course management, attendance tracking, marks and grading, medical record handling, notices, and timetables. This SRS is intended for use by the project development team, the course coordinators evaluating the project, and any future maintainers of the system.

This revision (Version 2.1) has been checked against the official mini-project specification. Where the specification is explicit, this document states the requirement as confirmed. Where the specification is silent or ambiguous, this document does not invent an official rule; instead it labels supporting design decisions as derived, marks unconfirmed capabilities as proposed, or flags the item for confirmation in Section 8.

1.2 Scope

The system, referred to as the Faculty of Technology Management System (FTMS), is a desktop application built using Java (with a graphical user interface) and MySQL as the backend database, following a layered architecture (presentation, service/business logic, data access). The system supports four categories of users — Admin, Lecturer, Student (Undergraduate), and Technical Officer — each with a distinct set of permissions.

The system is in scope for:

- Creation and maintenance of user profiles for all four roles, including account activation status.
- Creation and maintenance of course details and course materials.
- Recording and viewing of undergraduate attendance (theory and practical) and medical submissions.
- Recording and viewing of undergraduate marks, grades, SGPA, and CGPA.
- Determining exam eligibility based on attendance and continuous assessment (CA) criteria.
- Publishing and viewing of notices.
- Creation and viewing of timetables.

The system is out of scope for: online/web-based access outside the faculty network, financial or fee management, hostel or accommodation management, and integration with external university-wide systems (e.g. central LMS), unless otherwise agreed with the course coordinator.

1.3 Definitions, Acronyms and Abbreviations

Term	Description
FTMS	Faculty of Technology Management System (the proposed system)
CA	Continuous Assessment (assignments, quizzes, mid-exams, etc.)
GPA	Grade Point Average
SGPA	Semester Grade Point Average
CGPA	Cumulative Grade Point Average
UGC	University Grants Commission
Admin	Faculty administrator user role

Term	Description
TO	Technical Officer user role (used interchangeably with 'Technical Officer')
Undergraduate / Student	Student user role (used interchangeably throughout this document)
Theory (T) Session	A 2-hour lecture session; one session for one theory credit (1T) is 2 hours.
Practical (P) Session	A 2-hour practical/lab session; one session for one practical credit (1P) is 2 hours.
GUI	Graphical User Interface
DAO	Data Access Object
TBC	To Be Confirmed with the lecturer/course coordinator before implementation

1.4 References

- ICT2132 – Object Oriented Programming Practicum, Mini Project specification, Department of ICT, University of Ruhuna (primary source of truth for this document).
- UGC Commission Circular No. 12-2024 – Grading scheme for undergraduate degree programmes (exact grade boundaries/points to be sourced and confirmed; see Section 8).
- Current semester academic timetable of the Faculty of Technology (for course modules, credits and evaluation criteria).
- IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications (used as the structural template for this document).

1.5 Overview

Section 2 gives an overall description of the system, its user classes, and constraints, taken directly from the roles and permissions defined in the official specification. Section 3 describes functional requirements by module, each with a unique requirement ID for traceability. Section 4 covers external interface requirements. Section 5 specifies non-functional (quality) requirements. Section 6 covers data entities, relationships, and a requirements traceability matrix. Section 7 contains supporting appendices. Section 8 lists all points that remain open and require confirmation from the lecturer or course coordinator before implementation.

2. Overall Description

2.1 Product Perspective

FTMS is a new, standalone desktop software product developed independently for the Faculty of Technology. It is a self-contained three-layer application: a Java-based GUI (presentation layer), a business/service layer that implements all validation and calculation rules, and a MySQL relational database (data layer) accessed through JDBC.

2.2 Product Functions

Module	Summary of function
User Profile Management	Create, update, activate/deactivate, and view profiles for Admin, Lecturer, Student, and Technical Officer, with role-based edit restrictions.
Course & Material Management	Create and maintain course details, lecturer-course assignments, course materials, and course evaluation structures; ownership of Enrollment and assessment-criteria configuration is TBC (Section 8).
Attendance Management	Record theory/practical attendance per session and compute attendance percentages (theory, practical, combined).

Module	Summary of function
Medical Management	Record, approve/reject, and view medical submissions linked to specific student absence/attendance records.
Marks Management	Record CA and final exam marks against defined assessment components, out of 100.
Eligibility Checking	Determine CA eligibility and combined (attendance + CA) eligibility, for individuals and the whole batch.
Grading & GPA	Compute grades per subject using a configurable grade table, and compute SGPA/CGPA.
Notices	Create, publish, and view notices.
Timetable Management	Create and view class timetables, filtered by department, course, or role.

2.3 User Classes and Characteristics

The permissions below follow the official mini-project specification. Where an implementation-supporting scope restriction is required (for example, limiting a Lecturer to courses assigned to that Lecturer), it is treated as a derived design requirement in Section 3. Proposed capabilities are separately identified and listed in Section 8.

2.3.1 Admin

Administrative access to the four areas explicitly specified below. The Admin is assumed to be a non-technical or semi-technical faculty administrative staff member.

- Create and maintain all user profiles (Admin, Lecturer, Student, Technical Officer).
- Create and maintain courses.
- Create and maintain notices.
- Create and maintain timetables.

Note: the specification does not grant Admin direct access to marks, grades, GPA, attendance, or medical records. Any such access is treated as a Proposed capability requiring confirmation (see Section 8), not a confirmed requirement.

2.3.2 Lecturer

Academic staff member responsible for teaching one or more courses. Assignment-based access restrictions are treated as derived design requirements in Section 3. The Lecturer is assumed to have basic computer literacy.

- Update own profile, excluding username and password.
- Modify and add materials to courses assigned to them.
- Upload marks for all kinds of exams/components, for courses assigned to them.
- View undergraduate details.
- View undergraduate eligibility.
- View undergraduate marks, grades, and GPA.
- View attendance and medical records of undergraduates.
- View notices.

2.3.3 Technical Officer

Support staff responsible for maintaining attendance and medical records.

- Update own profile, excluding username and password.
- Add and maintain attendance details of undergraduates.

- Add and maintain medical details of undergraduates, including approval status.
- View notices.
- View timetables of their own department.

2.3.4 Undergraduate (Student)

End-user student with the most restricted permission set.

- Update only contact details and profile picture of their own profile.
- View attendance details.
- View medical details.
- View their own course details.
- View their own grades and GPA.
- View their own timetables.
- View notices.

Note: the specification does not explicitly state that Students may view their raw/final numeric marks (as distinct from grades and GPA). This is treated as a Proposed capability requiring confirmation (see Section 8), not a confirmed requirement.

2.4 Operating Environment

- Client: Windows/Linux desktop machine capable of running a Java Virtual Machine (JRE 17 or later recommended).
- Application: Java SE, using Swing or JavaFX for the GUI.
- Database: MySQL Server 8.x, accessed via JDBC.
- Deployment: standalone installation on faculty lab/office machines, sharing a common database server.

2.5 Design and Implementation Constraints

- The system must be implemented in Java and use MySQL as the database management system, as mandated by the course unit.
- The system must demonstrate classes and objects, inheritance, abstraction, polymorphism, encapsulation, exception handling, database handling, and a GUI.
- The system must store data for at least 1 Admin, 5 Lecturers, 4 Technical Officers, and 20 Undergraduates (including repeat/batch-missed students).
- Attendance must be recorded assuming 15 theory sessions and 15 practical sessions per course, with 1 credit for theory (1T) = one 2-hour session, and 1 credit for practical (1P) = one 2-hour session.
- CA marks must be out of 100, and a CA score of at least 40% is required for CA eligibility, as explicitly stated in the specification.
- Grades must be computed according to UGC Commission Circular No. 12-2024; the exact grade boundaries and grade points are not included in the specification and must be sourced and confirmed (see Section 8).

2.6 Assumptions and Dependencies

- Course modules, credit values, and CA evaluation criteria are taken from the current semester's academic timetable, as provided by course coordinators.
- Both theory and practical components of a course are assumed to have exactly 15 sessions each, regardless of credit value, as stated in the specification.

- The 80% figure used in the mandatory attendance test-data scenarios (Appendix A) is taken directly from the wording of the specification and is used only to define illustrative data bands. It is not assumed to be the official eligibility cutoff — see FR-ELG-02 and Section 8.
- The system is used within a single faculty and does not need to interoperate with other faculties' systems.
- A stable local network or single-machine deployment is available for MySQL connectivity.

3. Specific Requirements (Functional)

Each functional requirement is identified with a unique ID for traceability and mapped to the actor(s) who can perform it. Requirements marked "[Derived design requirement]" are implementation-supporting requirements inferred from the confirmed system behaviour but are not stated verbatim in the official specification. Requirements marked "[Proposed – requires confirmation]" are not explicitly stated in the official specification and must not be treated as confirmed until approved by the lecturer/course coordinator. Requirements marked "[Business rule TBC]" depend on a value or policy that is not stated in the specification; the associated calculation must not be hard-coded until that value is confirmed (see Section 8).

3.1 Authentication

ID	Requirement	Actor(s)
FR-AUTH-01	The system shall authenticate users via username and password before granting access, and route each user to a role-specific dashboard.	All
FR-AUTH-02	The system shall restrict every function to only the actor(s) explicitly authorized for it in this document, and shall deny access otherwise.	System

3.2 User Profile Management

ID	Requirement	Actor(s)
FR-USR-01	[Derived design requirement] The system shall allow the Admin to create and update user profiles for all four roles and to maintain an account status (active/inactive) used to control login access.	Admin
FR-USR-02	The system shall allow Lecturers and Technical Officers to update their own profile details, excluding username and password.	Lecturer, TO
FR-USR-03	The system shall allow Undergraduates to update only their contact details and profile picture.	Student
FR-USR-04	The system shall prevent any role from editing profile fields not permitted to that role (e.g., username, password, for non-Admin roles; account status, for any role other than Admin).	System
FR-USR-05	[Derived design requirement] The system shall allow Lecturers to view the profile details of undergraduates enrolled (via Enrollment) in courses assigned to that Lecturer (via Lecturer-Course Assignment).	Lecturer
FR-USR-06	[Derived design requirement] Every user account shall carry an account status (isActive / accountStatus) that determines whether the account can log in; only the Admin may change it.	Admin, System

3.3 Course and Material Management

ID	Requirement	Actor(s)
FR-CRS-01	The system shall allow the Admin to create and maintain course details, including course code, name, theory/practical credits, and semester.	Admin

ID	Requirement	Actor(s)
FR-CRS-02	[Derived design requirement] The system shall represent Lecturer–Course assignment explicitly so that permissions for materials, marks, and undergraduate views can be limited to courses assigned to the Lecturer.	Admin
FR-CRS-03	The system shall allow a Lecturer to add and modify materials only for courses they are currently assigned to.	Lecturer
FR-CRS-04	The system shall allow Students to view the details of courses they are enrolled in (via Enrollment).	Student

3.4 Attendance Management

ID	Requirement	Actor(s)
FR-ATT-01	The system shall allow Technical Officers to record attendance per enrolled student, per session, for both theory and practical components, across 15 sessions each.	TO
FR-ATT-02	The system shall compute attendance percentage separately for theory only, practical only, and combined, with the combined figure weighted by session hours (1T session = 2 hours, 1P session = 2 hours).	System
FR-ATT-03	The system shall display an attendance summary for the whole batch and for individual students.	Lecturer, TO
FR-ATT-04	[Derived design requirement] The system shall allow Lecturers to view attendance records of undergraduates enrolled in courses assigned to them and allow Technical Officers to view attendance records required for their attendance-maintenance duties.	Lecturer, TO
FR-ATT-05	The system shall allow Students to view their own attendance records only.	Student
FR-ATT-06	Seeded attendance data shall include, at minimum, the mandatory test-data scenarios defined in Appendix A (more than 80%, exactly 80%, less than 80% without medicals, more than 80% with medicals, and less than 80% with medicals), as required by the specification.	System (test data)

3.5 Medical Management

ID	Requirement	Actor(s)
FR-MED-01	[Derived design requirement] The system shall allow a Technical Officer to record a medical for a student and date and, where applicable, link that medical to one or more related absence/attendance records for that date. The record shall capture at minimum the student, medical date, supporting-document reference/path, status, and recording Technical Officer.	TO
FR-MED-02	[Derived design requirement] The system shall allow a Technical Officer to set and update the medical status as Pending, Approved, or Rejected, while preserving the specification requirement that approved status can be identified.	TO
FR-MED-03	[Proposed – requires confirmation] The system may allow a Student to submit or provide a supporting document reference for a Technical Officer to record; the Student shall not set or change the approval status.	Student, TO
FR-MED-04	The system shall allow Lecturers and Technical Officers to view medical records of undergraduates within the scope defined in FR-ATT-04, and allow Students to view only their own medical records.	Lecturer, TO, Student

Business Rule – To Be Confirmed with Lecturer/Course Coordinator: Whether an Approved medical (a) counts the session as attended, (b) is excluded from the total session count when calculating the percentage, or (c) has no effect on the attendance percentage and only affects eligibility directly, is not

stated in the specification. The calculation logic must not be implemented until this is confirmed. See Section 8.

3.6 Marks Management

ID	Requirement	Actor(s)
FR-MRK-01	The system shall allow a Lecturer to upload marks for all evaluation/assessment components of a course they are assigned to, following the course's evaluation criteria, with every component out of 100.	Lecturer
FR-MRK-02	[Derived design requirement] The system shall validate the configured weight percentages of a course's assessment components against the course coordinator's evaluation criteria before enabling final course mark calculation.	System
FR-MRK-03	The system shall display CA marks as a summary for the whole batch and for individuals.	Lecturer
FR-MRK-04	The system shall display final marks for individuals and for the whole batch.	Lecturer
FR-MRK-05	[Proposed – requires confirmation] The system may allow Students to view their own raw/final numeric marks, in addition to grade and GPA. This is not explicitly stated for the Undergraduate role in the specification.	Student (proposed)

3.7 Eligibility

ID	Requirement	Actor(s)
FR-ELG-01	The system shall determine CA eligibility as CA marks greater than or equal to 40%, as explicitly required by the specification.	System
FR-ELG-02	[Business rule TBC] The system shall determine attendance eligibility against an official attendance threshold, which is not stated in the specification and must be confirmed before implementation.	System
FR-ELG-03	The system shall determine overall exam eligibility as (attendance eligibility) AND (CA eligibility), for individuals and for the whole batch.	System
FR-ELG-04	The system shall allow Lecturers to view eligibility status (attendance, CA, and combined) of undergraduates individually and for the whole batch.	Lecturer

3.8 Grading and GPA

ID	Requirement	Actor(s)
FR-GPA-01	[Business rule TBC] The system shall compute a letter grade per subject using a configurable grade lookup table aligned to UGC Commission Circular No. 12-2024; the exact grade boundaries and grade points must be confirmed and loaded before go-live.	System
FR-GPA-02	[Business rule TBC] The system shall compute SGPA and CGPA for each undergraduate as a credit-weighted average of grade points; treatment of repeat, failed, and incomplete courses must follow official faculty/university regulations and must be confirmed before implementation.	System
FR-GPA-03	The system shall display marks and grades for each subject, for individuals, and for the whole batch.	Lecturer
FR-GPA-04	The system shall display SGPA and CGPA, with subjects and grades, for each undergraduate and for the whole batch. The whole-batch view shall list each student's own SGPA/CGPA (a per-student batch summary), not a single averaged figure, unless otherwise required by the lecturer.	Lecturer

ID	Requirement	Actor(s)
FR-GPA-05	The system shall allow Students to view their own grades and GPA.	Student

3.9 Notices

ID	Requirement	Actor(s)
FR-NOT-01	The system shall allow the Admin to create, update, and delete notices.	Admin
FR-NOT-02	The system shall allow all other roles (Lecturer, TO, Student) to view published notices.	Lecturer, TO, Student

3.10 Timetable Management

ID	Requirement	Actor(s)
FR-TIM-01	The system shall allow the Admin to create and maintain timetables for courses, including day, time, room, and session type.	Admin
FR-TIM-02	The system shall allow Technical Officers to view the timetable for their own department only.	TO
FR-TIM-03	The system shall allow Students to view their own course timetable.	Student

3.11 Proposed Capabilities Requiring Confirmation

The following capabilities are not explicitly stated in the specification but were considered by the project team as reasonable extensions. They must not be implemented as confirmed requirements until approved.

ID	Proposed capability	Actor(s)
FR-USR-07 (P)	Admin may optionally have read-only access to academic reports (attendance, marks, GPA, medicals) for institutional oversight purposes.	Admin (proposed)
FR-MRK-05 (P)	Students may view their own raw/final numeric marks, not only grades and GPA (see Section 3.6).	Student (proposed)
FR-ENR-01 (P)	The project may allow the Admin to create and maintain undergraduate course-enrollment records (Enrollment). The official specification uses course/student relationships but does not state who maintains enrollment data; the responsible actor must be confirmed.	Admin (proposed / TBC)
FR-ASM-01 (P)	The project shall store course assessment components and evaluation weights, but the actor authorized to configure them (e.g., Admin or assigned Lecturer) must be confirmed before this permission is implemented.	Admin / Lecturer (TBC)
FR-MED-03 (P)	Students may optionally provide a medical supporting-document reference for Technical Officer entry. Student submission is not explicitly granted in the official role description and therefore requires confirmation.	Student (proposed)

4. External Interface Requirements

4.1 User Interfaces

- A login screen accepting username and password, common to all roles.
- Role-specific dashboards: Admin dashboard (user, course, notice, timetable management), Lecturer dashboard (materials, marks, undergraduate views), Technical Officer dashboard (attendance, medical entry), Student dashboard (personal academic record).

- Forms for data entry (e.g., attendance sheet per session, marks entry sheet per component) with input validation and clear error messages.
- Report/summary screens with filter options (by student, by batch, by subject, by session type).

4.2 Hardware Interfaces

No special hardware interfaces are required. The system runs on standard desktop or laptop hardware capable of running a JVM.

4.3 Software Interfaces

- Java Runtime Environment (JRE) 17 or later.
- MySQL Server 8.x with the MySQL Connector/J JDBC driver.
- Swing or JavaFX libraries for the graphical user interface.

4.4 Communication Interfaces

The application communicates with the MySQL database server over a local network connection (TCP/IP) using JDBC. No external network or internet connectivity is required for core functionality.

5. Non-Functional Requirements

The specification does not itself prescribe non-functional/quality targets. The items below are quality requirements adopted by the project team as realistic, testable engineering practice for this system, not mandates from the official specification.

5.1 Performance (team-adopted target)

- The system shall aim to generate an individual attendance, marks, or eligibility report within approximately 3 seconds under normal load, for the minimum data volumes in Section 2.5.
- The system shall aim to generate a whole-batch summary report (20+ students) within approximately 5 seconds under normal load.

5.2 Security (team-adopted requirement)

- User passwords shall be stored as a password hash (passwordHash), never in plain text.
- Access to functionality shall be restricted according to the role-based permissions defined in Sections 2.3 and 3.
- Only Admin users shall be able to create user accounts or change account status.
- Students shall not be able to view or modify another student's records.

5.3 Reliability

- The system shall validate all data entry (e.g., marks out of 100, valid date ranges for medicals, assessment weights) before committing to the database.
- The system shall handle invalid input and database errors gracefully using exception handling, without crashing.

5.4 Usability

- The GUI shall use consistent navigation and terminology across all role dashboards.
- Error and confirmation messages shall be clear and in plain language suitable for non-technical staff.

5.5 Maintainability

- The system shall be implemented using a layered architecture (presentation, service/business logic, data access) to isolate changes.
- Database access shall be centralized in Data Access Object (DAO) classes to simplify future schema changes.
- Grade boundaries/points and any other values marked TBC in this document shall be stored as configurable lookup data, not hard-coded, so that confirmed values can be loaded without a code change.

6. Data Requirements, Use Cases, and Traceability

6.1 Key Data Entities

Entity	Key attributes (PK/FK noted)
User (abstract base)	userId (PK), username, passwordHash, fullName, contactNo, email, role, accountStatus (isActive)
Admin	extends User; no additional attributes required
Lecturer	extends User; department, designation
TechnicalOfficer	extends User; department
Undergraduate (Student)	extends User; registrationNo, batch, academicYear, studentStatus (REGULAR / REPEAT / BATCH_MISSED)
Course	courseId (PK), courseCode, courseName, theoryCredits, practicalCredits, semester
LecturerCourseAssignment	assignmentId (PK), lecturerId (FK → Lecturer), courseId (FK → Course), semester, academicYear
Enrollment	enrollmentId (PK), studentId (FK → Undergraduate), courseId (FK → Course), batch, semester, academicYear, attemptNo
CourseMaterial	materialId (PK), courseId (FK → Course), title, filePath/URL, uploadedBy (FK → Lecturer), uploadedAt
Assessment / MarkComponent	componentId (PK), courseId (FK → Course), componentName, componentType (CA / FINAL), weightPercentage, maximumMark (out of 100 basis), evaluationCriteria
Mark	markId (PK), studentId (FK → Undergraduate), componentId (FK → Assessment), marksObtained
AttendanceSession	sessionId (PK), courseId (FK → Course), sessionType (T/P), sessionNumber (1–15), date
AttendanceRecord	recordId (PK), studentId (FK → Undergraduate), sessionId (FK → AttendanceSession), status (Present/Absent), medicalId (FK → Medical, nullable; the same medical may link to multiple attendance records for the covered date)
Medical	medicalId (PK), studentId (FK → Undergraduate), medicalDate, supportingDocumentPath, status (Pending/Approved/Rejected), recordedBy (FK → TechnicalOfficer)
GradeRule	gradeRuleId (PK), minMark, maxMark, gradeLetter, gradePoint — configurable lookup table; values TBC (Section 8)
CourseResult	resultId (PK), studentId (FK → Undergraduate), courseId (FK → Course), finalMark (derived), grade (derived via GradeRule), gradePoint (derived)
Notice	noticeId (PK), title, body, postedBy (FK → Admin), datePosted, targetRole
Timetable	timetableId (PK), courseId (FK → Course), day, startTime, endTime, room, sessionType, department

Entity	Key attributes (PK/FK noted)
SemesterResult / GPA Summary	semesterResultId (PK), studentId (FK → Undergraduate), semester, academicYear, sgpa (derived); CGPA is derived cumulatively from eligible course/semester results and may be calculated rather than stored. Exact treatment rules are TBC in Section 8.

6.2 Use Case Diagram and Required Diagram Deliverables

Per the official mini-project specification, the required diagram deliverables (due 20 September 2026) are a UML class diagram and an ER diagram for the database. A use case diagram is not listed as a required deliverable in the specification; the project team may optionally prepare one as supporting documentation, but it must not be presented as a mandatory submission item. At a high level, each of the four actors (Admin, Lecturer, Technical Officer, Undergraduate) interacts with the system through the functions summarized in Section 2.3 and detailed in Section 3.

6.3 Requirements Traceability Matrix

This matrix maps each confirmed, derived, and proposed functional requirement to its primary actor, the data entity it operates on, and the module it belongs to, to support later development of the class diagram, ER diagram, implementation plan, and test cases.

ID	Actor	Related Entity	Related Module
FR-AUTH-01	All	User	Authentication
FR-AUTH-02	System	User	Authentication
FR-USR-01	Admin	User	User Profile Mgmt
FR-USR-02	Lecturer, TO	User	User Profile Mgmt
FR-USR-03	Student	User	User Profile Mgmt
FR-USR-04	System	User	User Profile Mgmt
FR-USR-05	Lecturer	Enrollment, LecturerCourseAssignment	User Profile Mgmt
FR-USR-06	Admin, System	User	User Profile Mgmt
FR-CRS-01	Admin	Course	Course Mgmt
FR-CRS-02	Admin	LecturerCourseAssignment	Course Mgmt
FR-CRS-03	Lecturer	CourseMaterial	Course Mgmt
FR-CRS-04	Student	Enrollment, Course	Course Mgmt
FR-ATT-01	TO	AttendanceSession, AttendanceRecord	Attendance
FR-ATT-02	System	AttendanceRecord	Attendance
FR-ATT-03	Lecturer, TO	AttendanceRecord, Enrollment	Attendance
FR-ATT-04	Lecturer, TO	AttendanceRecord, Enrollment	Attendance
FR-ATT-05	Student	AttendanceRecord	Attendance
FR-ATT-06	System	AttendanceRecord, Medical	Attendance
FR-MED-01	TO	Medical, AttendanceRecord	Medical
FR-MED-02	TO	Medical	Medical
FR-MED-03	Student, TO	Medical	Medical
FR-MED-04	Lecturer, TO, Student	Medical	Medical

ID	Actor	Related Entity	Related Module
FR-MRK-01	Lecturer	Mark, Assessment	Marks
FR-MRK-02	System	Assessment	Marks
FR-MRK-03	Lecturer	Mark, Assessment	Marks
FR-MRK-04	Lecturer	Mark, Assessment	Marks
FR-MRK-05 (P)	Student	Mark	Marks
FR-ELG-01	System	Mark, Assessment	Eligibility
FR-ELG-02	System	AttendanceRecord	Eligibility
FR-ELG-03	System	AttendanceRecord, Mark	Eligibility
FR-ELG-04	Lecturer	AttendanceRecord, Mark	Eligibility
FR-GPA-01	System	GradeRule, CourseResult	Grading & GPA
FR-GPA-02	System	CourseResult, SemesterResult	Grading & GPA
FR-GPA-03	Lecturer	Mark, CourseResult	Grading & GPA
FR-GPA-04	Lecturer	CourseResult, SemesterResult	Grading & GPA
FR-GPA-05	Student	CourseResult, SemesterResult	Grading & GPA
FR-NOT-01	Admin	Notice	Notices
FR-NOT-02	Lecturer, TO, Student	Notice	Notices
FR-TIM-01	Admin	Timetable	Timetable
FR-TIM-02	TO	Timetable	Timetable
FR-TIM-03	Student	Timetable	Timetable
FR-ENR-01 (P)	Admin (proposed / TBC)	Enrollment	Course / Enrollment Mgmt
FR-ASM-01 (P)	Admin / Lecturer (TBC)	Assessment	Course / Marks Mgmt

7. Appendices

Appendix A: Mandatory Attendance Test-Data Scenarios

The seeded attendance data must include at least one student for each of the following scenarios, corresponding to the scenarios specified in the official mini-project specification. These bands are used only to guarantee test-data coverage and are not, by themselves, a statement of the official eligibility cutoff (see FR-ELG-02 and Section 8).

Scenario	Description
A1	Undergraduate with more than 80% combined attendance.
A2	Undergraduate with exactly 80% combined attendance.
A3	Undergraduate with less than 80% combined attendance and no medicals.
A4	Undergraduate with more than 80% combined attendance and at least one medical on record.
A5	Undergraduate with less than 80% combined attendance and at least one medical on record.

Appendix B: Minimum Data Volume

- 1 Admin user.
- At least 5 Lecturer users.
- At least 4 Technical Officer users.
- At least 20 Undergraduate users (including repeat and batch-missed students).

Appendix C: Grading Reference (Pending Confirmation)

Final letter grades and grade points must be computed strictly according to UGC Commission Circular No. 12-2024. The exact grade boundaries and grade-point values are not included in the mini-project specification and are not stated here. They must be obtained from the current circular text, confirmed with the course coordinator, and loaded into the GradeRule lookup table (Section 6.1) before any grade is computed. No boundary values should be hard-coded in advance of that confirmation.

8. Items Requiring Lecturer / Course Coordinator Confirmation

The following points are not resolved by the official mini-project specification. No value has been assumed for any of them in the functional requirements above beyond what is needed to define test-data scenarios (Appendix A). They must be confirmed with the lecturer/course coordinator before the corresponding logic is implemented.

- The official attendance eligibility threshold (the specification requires attendance to factor into eligibility but does not state the exact required percentage).
- The treatment of an Approved medical in the attendance calculation — whether it (a) counts the session as attended, (b) is excluded from the total session count, or (c) affects eligibility only, without changing the displayed attendance percentage.
- The official grade boundaries and grade points defined in UGC Commission Circular No. 12-2024.
- The exact SGPA/CGPA calculation rules, including how repeat courses, failed courses, and incomplete courses are treated.
- Course-specific evaluation criteria and assessment component weights for each module (to be supplied by each course coordinator).
- Whether Students may view their own raw/final numeric marks, in addition to grades and GPA.
- Whether the Admin role may view academic reports (attendance, marks, GPA, medicals) for oversight purposes, beyond the four areas explicitly listed in the specification.
- Who is responsible for creating and maintaining Enrollment records that associate Undergraduates with courses (e.g., Admin or another authorized role).
- Which actor is authorized to configure each course's Assessment/MarkComponent definitions and evaluation weights (e.g., Admin, assigned Lecturer, or another authorized role).
- Whether Students are permitted to submit or provide medical supporting-document references directly for Technical Officer processing; this capability is proposed but not explicitly listed in the Undergraduate role description.